Lab 1

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Importance and Context

Our research at Rmadillos Consulting Group aims to address the question, "Do Democratic and Republican voters have different views of primary and secondary schooling?" This research is relevant as changing political climates have led to increased scrutiny in the media about what should and should not be taught in primary and secondary schools. Additionally, there has been continued debate about the importance and potential bias introduced in secondary education by both the Democrat and Republican parties. Our research plans to approach this question by analyzing sentiments regarding primary and secondary education in a non-leading or politically charged manner to gather true sentiments about these institutions across party lines. Our research aims to provide insights for policymakers to help inform policy decisions surrounding education and gain insights for future research that can be done on how to utilize these sentiments. Additionally, our findings can be used to raise awareness to the public about whether a significant difference in sentiments towards primary and secondary schooling actually exists between the two parties.

Data and Methodology

Our data comes from the American National Election Studies (ANES) - an academically-run national survey series that has collected data on public opinion, political participation, and voter behavior in the U.S. since 1948. We used the 2024 Pilot Study dataset, which is a cross-sectional survey that gathers public opinion during the early phase of the 2024 presidential election campaign. There are 1500 survey respondents resulting in 1909 in the data set due to weighting and partial responses in the survey techniques.

To answer our research question, we are interested in the sentiments of Democratic and Republican voters, as such we define voters as anyone who is registered to vote. We chose to look at registered voters because it is a more accurate measure of who may vote in future elections and considers factors such as turning 18, gaining citizenship or personal factors that may have affected voter participation in the previous election cycle. We analyzed the data and filtered out 72 respondents that responded "No", "Don't know" or did not answer regarding their voter registration status under `votereg`.

Research conducted by John Richard Petrocik found that many people have a party preference even if they do not choose to identify with that party. These people tend to have a political "leaning" and typically act as weak partisans despite identifying as an Independent or non-partisan. As such, if someone identified as a supporter of a party on the `pid7` variable, which measures political party identification on a 7-point scale, ranging from "Strong Democrat" to "Strong Republican" we categorized them as a Democrat or Republican accordingly. For the remaining respondents who identified as Independent, Not Sure or No Response we referred to the variable `pid_partylean` to find their party leaning and categorized them as the party they lean most closely with. All 333 respondents that did not identify with the Democrat or Republican party and showed no party leaning were considered true non-partisans and dropped from the data set.

To investigate if Republicans and Democrats have different views of primary education we chose to assess the variable `imp_schteach` which ranks how important they answered,"What is being taught in public schools?" on a likert scale from "Not at all important" to "Extremely important". We chose this variable to measure how partisan voters feel about primary education, which we consider to be K-12th grade public schools. In total we analyzed 1394 responses and dropped 110 partisans that did not respond to this question.

To evaluate if Democrats and Republicans have different views towards secondary education, we selected the `group_colprofs` variable to measure how respondents from each party answered the question, "How would you rate college professors?" which we use to assess sentiments towards secondary education, which we consider to be college and university levels of education. We analyzed 1361 responses and dropped 143 partisans that did not respond to this question.

We also considered assessing variables such as `highered_bias` (Political bias in what colleges/universities teach students) and `own_schteach` (assessing which party does a better job at what is being taught in school), but we did not want to introduce additional political bias. Other variables considered for both primary and secondary education had high non-response rates, which made them unsuitable for this analysis.

Table 1: Summary of Rows Remaining and Rows Dropped

Step	Rows_Remaining	Rows_Dropped
Initial Dataset (Total Rows)	1909	NA
After Voter Registration (votereg) Filter with Yes/No only	1837	72
After Filtering Party to only Democrats and Republicans	1504	333
After Dropping imp_schteach with inapplicable, legitimate skip	1394	110
After Dropping group_colprofs with not asked/ No Answer	1361	143

Testing the Importance of What is Being Taught in Public Schools

In order to test if there is a difference in sentiments between Democratic and Republican voters about the importance of what is being taught in public schools we conducted a Wilcoxon Rank Sum test using the Hypothesis of Comparisons. We determined this test to be the most appropriate for comparing these two groups because it allows us to compare the median ranks for subjective data between two independent groups. Additionally, the non-parametric nature does not rely on an assumption of a normal distribution (which cannot be made with non-metric data). We proceeded with this test based on the following assumptions:

The first assumption is an ordinal scale. As we can see, the importance of what is taught in public schools is ranked on a likert scale and there is a clear order to the data and it is not on a metric/ numerical scale. The second assumption is IID, where Republicans and Democrats were both drawn from the same distribution of ANES survey respondents and their selection is mutually independent. The ANES survey methodology also randomizes the order in which questions are asked, the order of the available responses and the questions that were asked of each participant which has the carry over effect of reducing bias in the aggregation of responses, supporting the notion of IID. We acknowledge that there may be challenges with this assumption due to temporal effects of current media coverage or other demographic and social factors not measured in this research.

The hypothesis of our Wilcoxon Rank Sum test can be phrased as follows:

Null Hypothesis (H_0): There is no difference in how Democratic (D) and Republican (R) voters rate the importance of what is being taught in public schools. $H_0 : P(D > R) = P(R > D) = 0.5$.
Alternative Hypothesis (H_1): There is a difference in how Democratic and Republican voters rate the importance of what is being taught in public schools. $H_1 : P(D > R) \neq 0.5$.

Testing the Sentiment Towards College Professors

In order to test if the sentiments towards college professors is significantly different between Republican and Democratic voters we conducted a two sample Welch's t-test as the thermometer scale data allows us to compare the mean sentiments of these two groups. We conducted a Levene Test which returned an F value of 77.114 indicating the variances between these two groups are not equal, making the Welch's t-test appropriate. We proceeded with this test based on the following assumptions:

The first assumption is metric scale. As the data is measured on a continuous numerical scale from 0-100, the data is both metric and ordinal. This allows us to assume metric scale because the data does not take on an exclusively ordinal structure. Our second assumption of IID is based on the same rationale described in detail above. ANES data is often assumed to be IID based on their weighted sampling methodology aiming to be as nationally representative as possible. Similarly recent media coverage or demographic, social or geographic variables not assessed in this study could potentially challenge our assumptions. Our third assumption of no major deviations from a normal distribution relies on the Central Limit theorem which states that with a large enough sample size the distribution will converge to a normal distribution. As our sample size is over 30 respondents this is considered a sufficiently large sample size under the CLT.

The hypothesis of our Welch's t-test can be phrased as follows:

Null Hypothesis (H_0): There is no difference between the mean sentiment toward college professors among Democratic and Republican voters. $H_0 : \mu_D = \mu_R$.
Alternative Hypothesis (H_1): There is a difference between the mean sentiment toward college professors among Democratic and Republican voters. $H_1 : \mu_D \neq \mu_R$.

For both tests conducted, if any of the above mentioned assumptions do not hold the statistical implications could result in inaccurate measures of the mean or median, incorrect or misleading conclusions and invalid results.

Results

The stacked bar chart (`Figure 1`) highlights the full distribution of importance ratings within each party, allowing us to compare how much weight each group assigns to public school teachings. Among Democratic voters, the largest share (33.93%) rated public school teachings as "4", while the most common rating among Republicans was "5" (60.36%). The Wilcoxon Rank-Sum test reveals statistically significant differences in the perceived importance of public school education between Democratic and Republican voters (p 0.001). Republicans demonstrated a stronger emphasis on what is being taught in public school, as evidenced by their substantially larger W statistic and a higher median (W = 317830.0, median = 5) compared to Democrats (W = 167018.0, median = 4). The rank-biserial correlation, a measure of effect size, was 0.31 (95% CI [0.26,0.36]), indicating a moderate practical difference between groups. This suggests Republicans prioritize school curriculum about 0.52 standard deviations more intensely than Democrats, a meaningful gap in the context of educational policy debates.

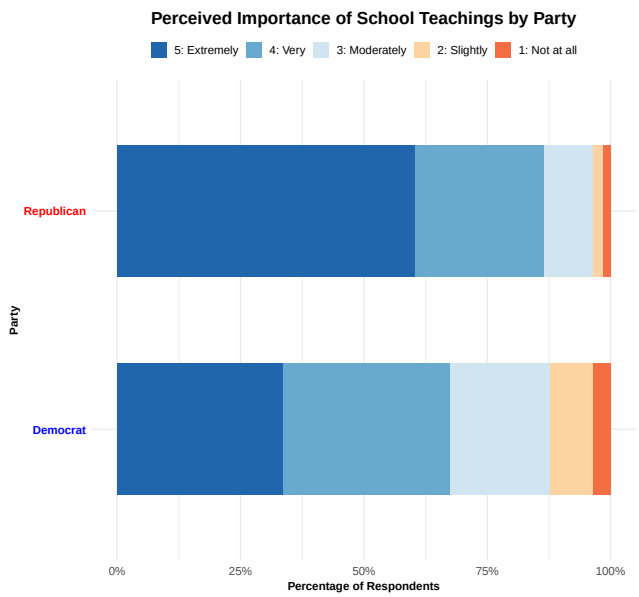

Figure 1: Relative Importance of Public School Teachings Among Democratic and Republican Voters

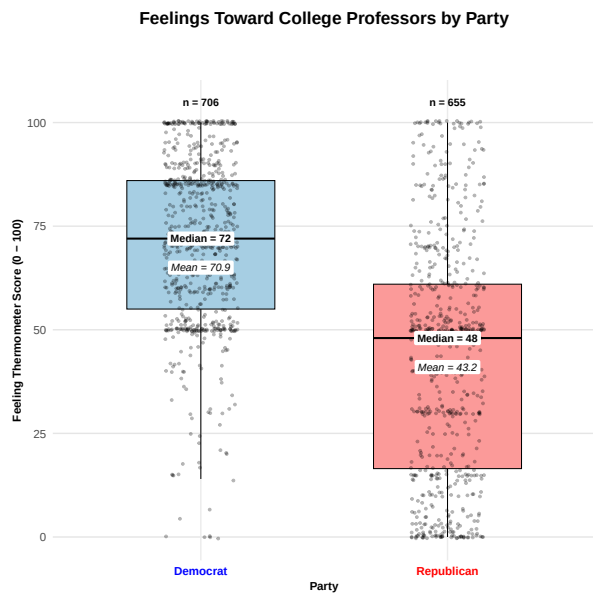

Figure 2: Distribution of Feelings Toward College Professors Among Democratic and Republican Voters

The box plot (`Figure 2`) highlights both the central tendency and the spread of responses, allowing us to see not only whether Democrats and Republicans feel differently about college professors, but also whether there is greater consistency or polarization within each party. The Welch's t-test reveals statistically significant differences in feelings toward college professors between Democratic and Republican voters (t = 20.43, p = 0.001). Democrats (mean = 70.9) rated college professors markedly higher than Republicans (mean = 43.2) with a very large effect size as measured by Cohen's d (delta = 0.97), reflecting a profound partisan divide. The direction of the effect states that Democrats hold substantially warmer sentiments toward college professors.

Discussion

This study identifies partisan divides in attitudes towards primary and secondary education. While Republicans place slightly greater importance on what is being taught in public schools, Democrats express significantly warmer feelings toward professors. Our results have practical significance for policymakers and educators seeking to understand how public sentiment in education institutions varies by political affiliation. The moderate effect size for primary school education suggests having both parties to support K-12 education is possible, whereas the polarization around professors implies higher education policies may face strong opposition. While this study addresses attitudes towards education in general, future studies may be focused on exploring the causal mechanisms– such as media influence, geographic background–behind the differences. Having a better understanding of how these attitudes influence policy preferences may help inform discussions about education and foster trust in educational institutions across different partisans.

Sources:

John Richard Petrocik,Measuring party support: Leaners are not independents, Electoral Studies, Volume 28, Issue 4, 2009,Pages 562-572,ISSN 0261-3794, https://doi.org/10.1016/j.electstud.2009.05.022.